

Tao 7

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1 Absolute Convergence

Let $\sum_{i=1}^{\infty} a_i$ be a convergent series of nonnegative numbers. Let $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ be a bijection. Then $\sum_{i=1}^{\infty} a_{f(i)}$ converges to the same number.

Proof: By ALT, it suffices to show the sequence $(\sum_{i=1}^n a_i - \sum_{i=1}^n a_{f(i)})_{n=1}^{\infty}$ converges to 0. Let $\epsilon > 0$. By Cauchy tail test and that a_i is nonnegative, pick $N_1 \in \mathbb{Z}^+$ such that $p, q \geq N_1 \rightarrow \sum_{i=p}^q a_i < \epsilon/2$. Let $S = f^{-1}([N_1])$. Since f is a bijection and $[N_1] \neq \emptyset$ is finite, S is nonempty finite. Hence, let $N_2 = \max S$.

Suppose $n \geq N_2$. Observe that (particularly for subsets of $\mathbb{Z}^+$), $S \subset [\max(S)] = [N_2] \subset [n]$. Hence, we have:

$$\left| \sum_{i=1}^n a_i - \sum_{i=1}^n a_{f(i)} \right| = \left| \sum_{i=1}^{N_1} a_i + \sum_{i=N_1+1}^n a_i - \sum_{i \in [S]} a_{f(i)} - \sum_{i \in [n]-S} a_{f(i)} \right|.$$

Since f is a bijection and by definition of S , f restricts to a bijection $g : S \rightarrow [N_1]$. By replacement, $\sum_{i \in [S]} a_{f(i)} = \sum_{i \in [N_1]} a_i = \sum_{i=1}^{N_1} a_i$.

Since $[n] \neq \emptyset$ is finite, $f([n]) \neq \emptyset$ is finite. Let $M = \max f([n])$. If $i \in [n] - S$, then $f(i) \leq M$. Further, since f is a bijection, $f(i) \notin f(S) = [N_1]$, so $f(i) \geq N_1 + 1$. It follows $f(i) \in \{k \in \mathbb{Z}^+ : N_1 + 1 \leq k \leq M\} = A$. Since f is injective, f restricts to a bijection of $[n] - S$ with a subset B of A . By replacement, we may sum over B .

Now we finish:

$$\begin{aligned}
\left| \sum_{i=1}^n a_i - \sum_{i=1}^n a_{f(i)} \right| &= \left| \sum_{i=1}^{N_1} a_i + \sum_{i=N_1+1}^n a_i - \sum_{i=1}^{N_1} a_i - \sum_{i \in B} a_i \right| \\
&\leq \sum_{i=N_1+1}^n a_i + \sum_{i \in B} a_i \leq \sum_{i=N_1+1}^n a_i + \sum_{i \in A} a_i \\
&= \sum_{i=N_1+1}^n a_i + \sum_{i=N_1+1}^M a_i < \epsilon/2 + \epsilon/2 = \epsilon.
\end{aligned}$$

2 Root Test

Let (a_n) be a sequence such that $\limsup_{n \rightarrow \infty} |a_n|^{1/n} \rightarrow \alpha$. Then:

- (1). If $\alpha < 1$, then (a_n) converges absolutely.
- (2). If $\alpha > 1$, then (a_n) does not converge conditionally.
- (3). If $\alpha = 1$, then the test is inconclusive.

Suppose $\alpha < 1$. Let $\epsilon = (1-\alpha)/2 > 0$ and pick $N \in \mathbb{Z}^+$ such that $\sup_{m \geq N} |a_m|^{1/m} < \alpha + (1-\alpha)/2 = (1+\alpha)/2$ (by using $\alpha = \inf_{M \geq 1} \sup_{n \geq M} |a_n|^{1/n}$). Let $\alpha' = (1+\alpha)/2$.

For all $n \geq N$, we have $|a_n| \leq (\alpha')^n$. $\sum_{n=N}^{\infty} (\alpha')^n$ converges by Geometric series test. By Comparison test, (a_n) converges.

Now suppose $\alpha > 1$. Let $\alpha' = (1+\alpha)/2 \in (1, \alpha)$. Then for all $N \geq 1$ there exists $n \geq N$ such that $|a_n|^{1/n} > \alpha'$. Construct an increasing sequence $b : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ such that $|a_{b(n)}|^{1/b(n)} > \alpha'$, so $|a_{b(n)}| > (\alpha')^{b(n)} \geq (\alpha')^n$. Then the subsequence $(a_{b(n)})$ does not converge to 0, so the series $\sum_{n=1}^{\infty} a_n$ does not converge by zero test.

Now for the last case. The harmonic series $\sum_{n=1}^{\infty} 1/n$ does not converge and $(1/n)^{1/n} \rightarrow 1$ since $n^{1/n} \rightarrow 1$. The Basel sum $\sum_{n=1}^{\infty} 1/n^2$ converges (by say, p -series test) and $(1/n^2)^{1/n} = ((1/n)^{1/n})^2 \rightarrow 1$.

Let (x_n) be a sequence. Denote by $S = \{n \in \mathbb{N} : x_n \neq 0\}$. Suppose S is infinite and let $a : \mathbb{N} \rightarrow S$ be the enumeration. Then $\sum_{n=0}^{\infty} x_n$ converges iff $\sum_{n=0}^{\infty} x_{a(n)}$ converges.

Lemma: $\sum_{n=0}^{a(m)} x_n = \sum_{n=0}^m x_{a(n)}$.

Proof: $\sum_{n=0}^m x_{a(n)} = \sum_{n \in a([m])} x_n = \sum_{n \in [a(m)] \cap S} x_n = \sum_{n \in [a(m)] \cap S} x_n + \sum_{n \in [a(m)] - S} x_n = \sum_{n \in [a(m)]} x_n = \sum_{n=0}^{a(m)} x_n$, as desired.

Suppose $\sum_{n=0}^{\infty} x_n = L$. Let $\epsilon > 0$. Pick N such that $m \geq N \rightarrow |\sum_{n=0}^m x_n - L| < \epsilon$. Then for $m \geq M$, $a(m) \geq m \geq N$ and $|\sum_{n=0}^m x_{a(n)} - L| = |\sum_{n=0}^{a(m)} x_n - L| < \epsilon$, so $\sum_{n=0}^{\infty} x_{a(n)} = L$.

Fubini's theorem for infinite sums. Let $\sum_{(n,m) \in N^2} f(n,m)$ be absolutely convergent. Then $\sum_{n=0}^{\infty} \sum_{m=0}^{\infty} f(n,m) = \sum_{(n,m) \in N^2} f(n,m)$.

Let $g : N \rightarrow N^2$ be a bijection and let $\epsilon > 0$. Since $\sum_{(n,m) \in N^2} f(n,m)$ is absolutely convergent, pick $M \in N$ such that for $p, q \geq M$ we have $\sum_{k=p}^q |f(g(k))| < \epsilon$. Since $\pi_1 g([M]) \subset N$ is nonempty finite, let N' be its maximum element. Let $N' \geq N$. Let $g' : N \rightarrow N^2$ be the bijection such that for $k = (N' + 2)q + r$ (by division algorithm) and $r \leq N'$, we have $g'(k) = (r, q)$ and $g'' = g'|((N' + 2)N + N' + 1)$ bijects with $N_{\geq N'+1} \times N$.

Then

$$\begin{aligned} \sum_{k=0}^{\infty} f(g(k)) &= \sum_{k=0}^{\infty} f(g'(k)) = \sum_{q=0}^{\infty} \sum_{r=0}^{N'+1} f g'((N' + 2)q + r) \\ &= \sum_{q=0}^{\infty} \sum_{r=0}^{N'} f g'((N' + 2)q + r) + f g'((N' + 2)q + N' + 1) \\ &= \sum_{q=0}^{\infty} \sum_{r=0}^{N'} f(r, q) + f g'((N' + 2)q + N' + 1) \\ &= \sum_{r=0}^{N'} \sum_{q=0}^{\infty} f(r, q) + \sum_{q=0}^{\infty} g''(q). \end{aligned}$$

Note the last inequality uses the absolute convergence of $\sum_{(n,m) \in N^2} f(n,m)$. For any $i \in N$, since $\pi_1(g''(i)) > N' \geq N \rightarrow g''(q) \notin g([M])$, we have $|\sum_{q=0}^p g''(q)| \leq \sum_{q=0}^p |g''(q)| < \epsilon$, so $|\sum_{r=0}^{N'} \sum_{q=0}^{\infty} f(r, q) - \sum_{(n,m) \in N^2} f(n,m)| < \epsilon$ for $N' \geq N$. Since this holds for all $\epsilon > 0$, we have $\sum_{n=0}^{\infty} \sum_{m=0}^{\infty} f(n,m) = \sum_{(n,m) \in N^2} f(n,m)$.

3 Sums

Let X be finite and let $f : X \rightarrow \mathbb{R}$ be a function. Let $g : [n] \rightarrow X$ be a bijection and define $\sum_{x \in X} f(x) = \sum_{i=1}^n f(g(i))$.

Claim: $\sum_{x \in X} f(x)$ is well-defined.

Proof: We induct on n . For $n = 0$, let $g : \emptyset \rightarrow X$ be any bijection, so g must be the empty function. $\sum_{i=1}^0 f(g(i)) = 0$ is invariant.

Suppose the result holds for n and let $|X| = n + 1$. Let $g, h : [n + 1] \rightarrow X$ be bijections. Since h is surjective, there exists $i_0 \in [n + 1]$ such that $g(n + 1) = h(i_0)$. Let $\tilde{h} : [n + 1] \rightarrow X$ be the bijection such that $\tilde{h}(i) = h(i)$ for $i \neq i_0, n + 1$, $\tilde{h}(i_0) = h(n + 1)$ and $\tilde{h}(n + 1) = h(i_0) = g(n + 1)$, so $\tilde{h} : [n + 1] \rightarrow X$ is a bijection.

Since $\tilde{h}(n + 1) = g(n + 1) = x_0 \in X$, let $X' = X - x_0$, so $|X'| = n$. Let $\tilde{h}', g'$ be the restrictions of $\tilde{h}, g$ to $[n]$, so $\tilde{h}', g'$ are bijections of $[n]$ with X' . By the inductive hypothesis, $\sum_{i=1}^n f(\tilde{h}'(i)) = \sum_{i=1}^n f(g'(i))$. Now if $i_0 \neq n + 1$:

$$\begin{aligned} \sum_{i=1}^{n+1} f(h(i)) &= \sum_{i=1}^{i_0-1} f(h(i)) + f(h(i_0)) + \sum_{i=i_0+1}^n f(h(i)) + f(h(n+1)) \\ &= \sum_{i=1}^{i_0-1} f(\tilde{h}(i)) + f(\tilde{h}(i_0)) + \sum_{i=i_0+1}^n f(\tilde{h}(i)) + f(g(n+1)) = \sum_{i=1}^n f(\tilde{h}'(i)) + f(g(n+1)) \\ &= \sum_{i=1}^n f(g'(i)) + f(g(n+1)) = \sum_{i=1}^n f(g(i)) + f(g(n+1)) = \sum_{i=1}^{n+1} f(g(i)). \end{aligned}$$

Let $\sum_{n=0}^{\infty} x_n$ be absolutely convergent and (x_{a_n}) a subsequence. Then $\sum_{n=0}^{\infty} x_{a_n}$ is absolutely convergent.

Since $\sum_{n=0}^{\infty} |x_n|$ is absolutely convergent, the sequence of partial sums $\sum_{n=0}^m |x_n|$ is bounded. Then $\sum_{n=0}^m |x_{a(n)}| = \sum_{n \in a([m])} |x_n| \leq \sum_{n \in [a(m)]} |x_n|$ is bounded.

Suppose $X = A \cup B$ for A, B infinite countable disjoint. Let $a : N \rightarrow A$, $b : N \rightarrow B$ be bijections and let $g : N \rightarrow X$ by $g(n) = a(n/2)$ if n is even and $g(n) = b((n-1)/2)$ if n is odd, so $g : N \rightarrow X$ is a bijection.

Then $\sum_{x \in X} f(x) = \sum_{i=0}^{\infty} f(g(i)) = \sum_{i=0}^{\infty} f(g(2i)) + f(g(2i+1))$. Observe that $\sum_{i=0}^{\infty} f(g(2i)) = \sum_{i=0}^{\infty} f(a(i))$ is absolutely convergent as a subsequence of $f(g(i))$. Since $a : N \rightarrow A$ is a bijection, $\sum_{x \in A} f(x)$ is absolutely convergent and equal to $\sum_{i=0}^{\infty} f(a(i))$. Similar results hold for g , so $\sum_{x \in X} f(x) = \sum_{i=0}^{\infty} f(a(i)) + \sum_{i=0}^{\infty} f(b(i)) = \sum_{x \in A} f(x) + \sum_{x \in B} f(x)$.

Now suppose A is finite with cardinality n . Let $b : N \rightarrow B$ be a bijection.

Let $\tilde{b} : N \rightarrow X$ be a bijection such that $b|N(n)$ is a bijection with A and $\tilde{b}(i) = b(i - n)$ for $i \geq n$.

$$\text{Then } \sum_{x \in X} f(x) = \sum_{i=0}^{\infty} f(\tilde{b}(i)) = \sum_{i=0}^{n-1} f(\tilde{b}(i)) + \sum_{i=n}^{\infty} f(\tilde{b}(i)) = \sum_{i=0}^{n-1} f(\tilde{b}(i)) + \sum_{i=0}^{\infty} f(b(i)) = \sum_{x \in A} f(x) + \sum_{x \in B} f(x).$$

If A, B are finite, then the result is from before.

Let $g : N \rightarrow N^2$ be a bijection. Let $\epsilon > 0$. Since $\sum_{(n,m) \in N^2} f(n,m)$ is absolutely convergent, pick $M \in N$ such that $p, q \geq M \rightarrow \sum_{n=p}^q |f(g(n))| < \epsilon$. Pick $N \in N$ such that $(n,m) \notin [[N] \times N \rightarrow (n,m) = g(k)$ for some $k \geq M$. Then for $N' \geq N$, we have $\sum_{(n,m) \in N^2} f(n,m) = \sum_{(n,m) \in [[N'] \times N} f(n,m) + \sum_{(n,m) \in (N - [[N']] \times N} f(n,m)$. Let $h : N \rightarrow (N - [[N']] \times N$ be a bijection.

$$\text{Then for all } n \in N, |\sum_{i=0}^n f(h(i))| \leq \sum_{i=0}^n |f(h(i))| < \epsilon, \text{ so } |\sum_{(n,m) \in (N - [[N']] \times N} f(n,m)| = |\sum_{i=0}^{\infty} f(h(i))| \leq \epsilon.$$

I claim that $\sum_{(n,m) \in [[N']] \times N} f(n,m) = \sum_{n=0}^{N'} \sum_{m=0}^{\infty} f(n,m)$. We proceed by induction. The base case: $\sum_{(n,m) \in [[0] \times N} f(n,m) = \sum_{m \in N} f(0,m) = \sum_{m=0}^{\infty} f(0,m)$. Assume the result for N' . Then $\sum_{(n,m) \in [[N'+1] \times N} f(n,m) = \sum_{(n,m) \in [[N'] \times N} f(n,m) + \sum_{(n,m) \in \{N'+1\} \times N} f(n,m) = \sum_{n=0}^{N'} \sum_{m=0}^{\infty} f(n,m) + \sum_{m=0}^{\infty} f(N'+1,m) = \sum_{n=0}^{N'+1} \sum_{m=0}^{\infty} f(n,m)$, completing the induction.

Then for all $N' \geq N$ we have shown $|\sum_{n=0}^{N'} \sum_{m=0}^{\infty} f(n,m) - \sum_{(n,m) \in N^2} f(n,m)| \leq \epsilon$, so $\sum_{n=0}^{\infty} \sum_{m=0}^{\infty} f(n,m) = \sum_{(n,m) \in N^2} f(n,m)$.

To finish the theorem, let $f' : N^2 \rightarrow \mathbb{R}$ be given by $f'(m,n) = f(n,m)$. There is the natural bijection $b : N^2 \rightarrow N^2$ given by $(n,m) \mapsto (m,n)$. Then $\sum_{(n,m) \in N^2} f(n,m) = \sum_{(m,n) \in N^2} f(b(m,n)) = \sum_{(m,n) \in N^2} f(n,m) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} f(n,m)$.

Let X be countably infinite and $f : X \rightarrow \mathbb{R}$ a function. Then $\alpha = \sup\{\sum_{x \in A} |f(x)| : A \text{ is finite}\} < \infty$ iff $\sum_{x \in X} f(x)$ is absolutely convergent.

Suppose $\sum_{x \in X} f(x)$ is absolutely convergent. Then for $L = \sum_{x \in X} |f(x)|$ and any finite subset A , we have $\sum_{x \in A} f(x) \leq \sum_{x \in X} f(x)$, so $\alpha < \infty$.

Conversely, suppose $\alpha < \infty$. Let $g : N \rightarrow X$ be a bijection and take $A = g([n])$ for all $n \in N$. Then $\sum_{i=0}^n |f(g(i))| \leq \alpha$ and by monotone convergence theorem converges.

$\mathbb{R}$ is uncountable. Consider the function $f : P(N) \rightarrow \mathbb{R}$ given by $f(A) = \sum_{x \in A} 10^{-x}$. Since $\sum_{x \in N} 10^{-x} = \sum_{n=0}^{\infty} 10^{-n} = \frac{10}{9}$ is absolutely convergent and $A \subset N$, $\sum_{x \in A} f(x)$ is absolutely convergent, and in particular well-defined. It is enough to show that f is injective.

Suppose $A \neq B$ and $f(A) = f(B)$. Let $n \in \Delta(A, B)$ be minimal by Well-ordering of N . WLOG $n \in A$ and $n \notin B$. Then $X = A \cap [[n-1] = B \cap [[n-1]$, $n \in A - X$, $n \notin B - X$. Since $A, X - A$ are disjoint, we have

$$\sum_{x \in A} 10^{-x} = \sum_{x \in X} 10^{-x} + \sum_{x \in A-X} 10^{-x}.$$

By assumption,

$$\sum_{x \in A-X} 10^{-x} = \sum_{x \in B-X} 10^{-x}.$$

Observe that $B - X \subset \{n+1, n+2, \dots\}$, so

$$\sum_{x \in B-X} 10^{-x} \leq \sum_{k \geq n+1} 10^{-k} = 10^{-(n+1)} \cdot \frac{10}{9} = 10^{-n}/9.$$

On the other hand,

$$\sum_{x \in A-X} 10^{-x} \geq 10^{-n},$$

so $\sum_{x \in A-X} 10^{-x} > \sum_{x \in B-X} 10^{-x}$, absurd.